ELIAS AL ALAM

Environmental Engineer

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Swiss Work Permit B

EDUCATION

Master of Environmental Science: Hazards & Risk

University of Lausanne, Lausanne, Switzerland

08/2022 - 07/2025 | Switzerland

- Thesis: Investigated the impact of tributary inputs on sediment size distribution downstream of the Grande Dixence dam using high-resolution imagery and photogrammetry techniques.
- Developed a supervised learning model achieving 97% accuracy for detecting oil spills from satellite images.

B.E. Mechanical Engineering (ABET Accredited)

Lebanese American University, Byblos, Lebanon

08/2016 - 05/2021 | Lebanon

- Designed and built a cost-effective tribometer to test wear and friction of RZeppa joints, reducing manufacturing costs to 1/40 of commercial alternatives.
- Developed a smart home and automated irrigation system using LabVIEW, enhancing energy efficiency and reducing water consumption by 30%.

LANGUAGES

English Native ●●●●●

French Proficient ●●●●○

Arabic Native ●●●●●

German Intermediate ●●○○○

PROFESSIONAL EXPERIENCE

R&D Engineer & GIS Analyst

Inverto Earth

03/2026 - 10/2026

- Built and validated Bayesian hierarchical models in Stan for above-ground biomass estimation across mangrove ecosystems, with full posterior uncertainty quantification and spatial random effects to account for autocorrelation; selected Stan over PyMC and Turing.jl after benchmarking fit time and accuracy.
- Developed Google Earth Engine pipelines (JS and Python) for multi-source satellite data extraction across Landsat, Sentinel-2, and GEDI, with automated cloud-free image screening and monthly compositing over 3+ scenes to reduce sampling bias.
- Built a Flask web application for end-to-end analysis: CSV ingest, automated quality control and type flagging, model fitting, and visualized results with interpretable coefficient summaries.
- Working knowledge of carbon crediting standards and methodologies across Verra (VM0033 tidal wetland and seagrass restoration, VM0047 ARR), Gold Standard, Isometric, and OxCarbon, plus UNFCCC sampling and calculation tools
- Reviewed project design documents, GHG statements, and validation/verification reports against methodology requirements, and assessed rating-agency frameworks used to score project quality and permanence.

SKILLS

Geospatial & Remote Sensing

Google Earth Engine, GEDI, Sentinel, Landsat, QGIS, ArcGIS, lidR, xDEM, 3D modeling, LiDAR & UAV/SfM processing, Environmental Impact Assessment, Life Cycle Assessment, Sustainability Analysis

Programming & Development

Python, R, Stan, Julia, Rust, JavaScript, MATLAB, HTML/CSS, Flask, Docker, Git/GitLab, Linux

Machine Learning

scikit-learn, XGBoost, Random Forest, BART, NGBoost, AutoGluon, feature selection (mutual information, VIF), spatial cross-validation

Bayesian & Statistical Modeling

Stan, PyMC, Turing.jl, hierarchical models, MCMC/NUTS, variational inference, spatial statistics, uncertainty propagation, conformal prediction, model diagnostics (R-hat, ESS, Pareto-k)

Engineering & Simulation

Thermodynamics & Mechanics, Statics & Dynamics, Finite Element Analysis, Computational Fluid Dynamics, MATLAB, SolidWorks, AutoCAD, Ansys, Fluid Mechanics, Materials Science

Carbon MRV & Standards

Verra (VM0033, VM0047), Gold Standard, Isometric, OxCarbon, UNFCCC sampling & calculation tools, PDD and V&V report review, dMRV, GHG accounting, above-ground biomass estimation

SOCIAL MEDIA

Portfolio
https://www.eliasalalam.dev

LinkedIn
https://www.linkedin.com/in/elias-al-alam/

GitHub
https://github.com/michmazbout

PROFESSIONAL EXPERIENCE

Research Assistant (40%)

University of Lausanne, Lausanne, Switzerland

03/2023 - 09/2024

- Analyzed 12 years of glacial elevation data from 6 glaciers, identifying critical erosion points.
- Collaborated with researchers from EPFL, ETH Zurich, and Eawag on a multidisciplinary glacier study.
- Utilized QGIS, ArcGIS, and Python/XDEM to track changes in elevation and proglacial area.

Intern: Geophysical Field Engineer

Geo2x & GEOOL, Yverdon-les-Bains, Switzerland

08/2023 - 09/2023

- Deployed and monitored 5,000 seismic sensors across Switzerland for a national geothermal reservoir mapping project.
- Managed sensor installation and data collection under challenging conditions.

Intern: Mechanical Engineering Technician

DEMCO Steel, Beirut, Lebanon

06/2019 - 09/2019

- Performed inspections and diagnostics on industrial machinery, increasing production efficiency by 25% and reducing downtime by 150 hours per month.

REFERENCES

Available upon request